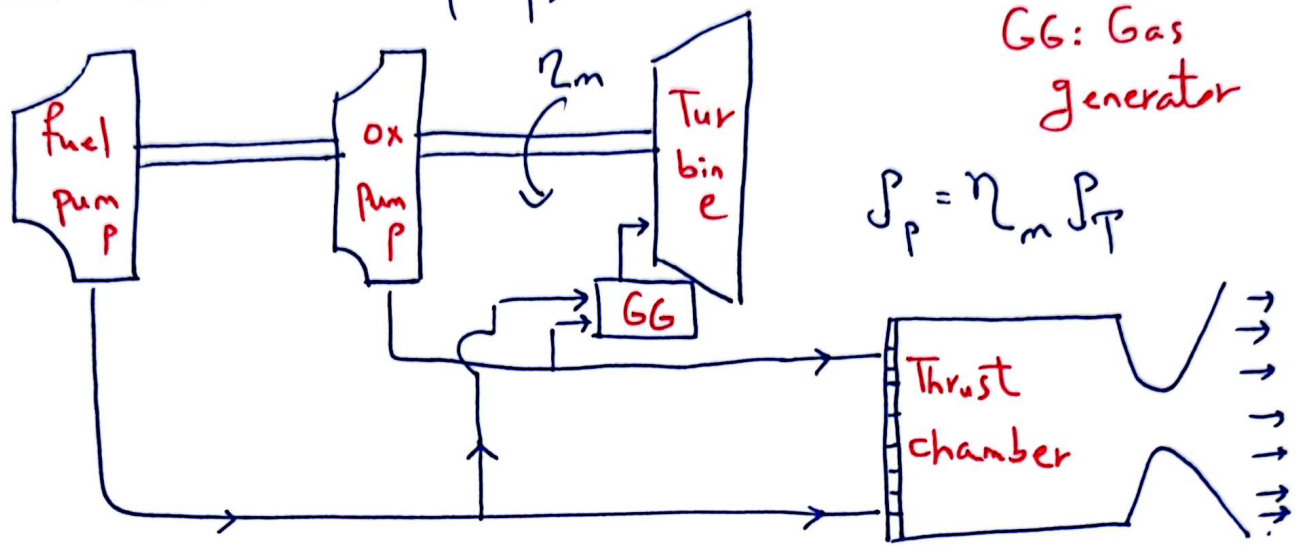


Fri-04-04-2025 LPRE feed system: Turbine - Solid rocket

In previous session, we considered pumps. Now, we consider turbine that rotates the pumps.



Turbine Analysis:

First law: $\dot{m}_T (h_{o2} - h_{o1}) = -\dot{P}_T$

$$h_{o2} - h_{o1} = \frac{-\dot{P}_T}{\dot{m}_T} = -w_{iT}$$

$$\text{or } w_{iT} = h_{o1} - h_{o2} = c_p (T_{o1} - T_{o2})$$

Ideal turbine is isentropic. Since the kinetic energy of the turbine exhaust is wasted, the turbine is designed to make it as small as possible, ideally zero. Thus:

$$w_{iT_s} = h_{o1} - h_{o2s} = c_p (T_{o1} - T_{o2s})$$

isentropic

$$h_{o2} = h_{o2s} + \frac{u_{o2}^2}{2}$$

(P-1)

We define turbine efficiency:

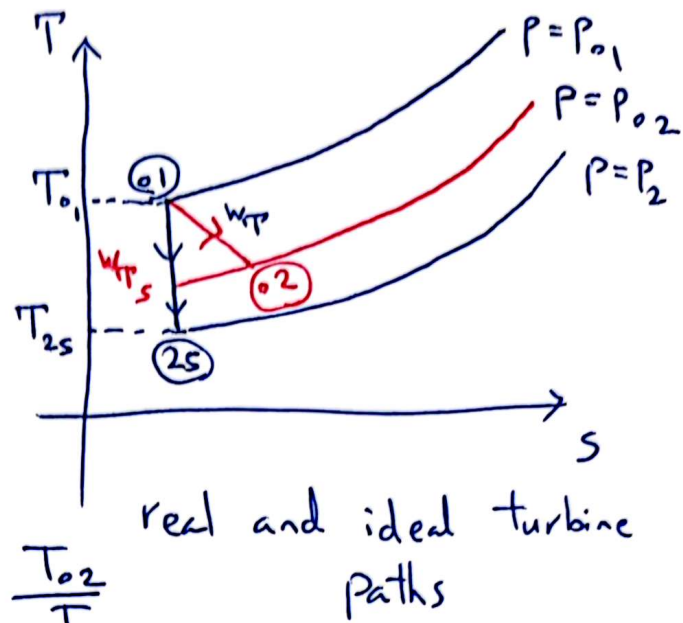
$$\eta_T = \frac{w_T}{w_{T_s}} = \frac{c_p (T_{01} - T_{02})}{c_p (T_{01} - T_{2s})}$$

$$= \frac{T_{01} \left(1 - \frac{T_{02}}{T_{01}}\right)}{T_{01} \left(1 - \frac{T_{2s}}{T_{01}}\right)}$$

$$= \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{P_{2s}}{P_{01}}\right)^{\frac{\gamma-1}{\gamma}}} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{P_2}{P_{01}}\right)^{\frac{\gamma-1}{\gamma}}}$$

We used isentropic relation

$$P_2 = P_{2s}$$



Summary of turbine Eq.:

$$\dot{Q}_T = \dot{m}_T w_T, \quad w_T = c_p (T_{01} - T_{02}), \quad \eta_{T_s} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{P_2}{P_{01}}\right)^{\frac{\gamma-1}{\gamma}}}$$

The turbine wheel is turned by the applied pressure ratio, $\frac{P_{01}}{P_{02}}$ across it. The turbine power is used to turn the fuel and oxidizer pumps. $\Rightarrow$

$$\dot{Q}_P = \eta_m \dot{Q}_T$$

"mechanical efficiency"

(P-2)

Example: A turbine, powered by gas generator, is used to derive the pumps of the previous example. ($P_p = 0.6 \text{ MW}$)

The turbine entry stagnation conditions are $\underbrace{2 \text{ MPa}}_{P_{01}}$ and $\underbrace{800 \text{ K}}_{T_{01}}$.

The turbine discharge static pressure is $\underbrace{0.14 \text{ MPa}}_{P_2}$.

calculate the fraction of the total propellant $\underbrace{P_2}_{\dot{m}_p}$ flow to be diverted to gas generator ($\frac{\dot{m}_p}{\dot{m}}$), $\bar{M} = 34$, $\gamma = 1.3$, $\eta_{T_s} = 70\%$
 $\underbrace{92 \frac{\text{kg}}{\text{s}}}_{\dot{m}} \text{ (previous Ex.)}$ $\eta_m = 90\%$

Solution: $R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{34} = 244.5 \frac{\text{J}}{\text{kg K}}$

$$c_p = \frac{\gamma R}{\gamma - 1} = 1060 \frac{\text{J}}{\text{kg K}}$$

We use these Eqs. $\dot{Q}_T = \dot{m}_T W_T = \dot{m}_T \eta_{T_s} W_{T_s} = \dot{m}_T c_p (T_{01} - T_2)$

To get T_{02} use η_{T_s} relation:

$$\eta_{T_s} = 0.7 = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{P_2}{P_{01}}\right)^{\frac{\gamma-1}{\gamma}}} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{0.14 \text{ M}}{2 \text{ M}}\right)^{\frac{1.3-1}{1.3}}}$$

Solve the above Eq.: $\Rightarrow \frac{T_{02}}{T_{01}} = 0.6790$
 $\underbrace{T_{01}}_{800 \text{ K}} = 800 \text{ K}$

$$\Rightarrow T_{02} = 0.6790 \times 800 = \boxed{543.2 \text{ K}}$$

(P-3)

$$\dot{Q}_T = \dot{m}_T c_p (T_{01} - T_{02}) = \dot{m}_T \times (1060) (800 - 543.2) = 272208 \dot{m}_T$$

now use this:

$$\eta_m \dot{Q}_T = \dot{Q}_p$$

$$0.9 \times 272208 \dot{m}_T = 6 \times 10^5 \implies \dot{m}_T = \boxed{2.45 \frac{\text{kg}}{\text{s}}}$$

$$\frac{\dot{m}_T}{\dot{m}} = \frac{2.45}{92} = 0.0266 \quad \text{or} \quad 2.66\%$$
